STA141A Fundamentals of Statistical Data Science

Final Project Report - Team 15

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I. Introduction.

This project seeks to analyze the Heart Disease Dataset to understand the factors influencing heart diseases and create predictive models for identifying at-risk individuals. With the rising global prevalence of heart diseases, there's a need for effective predictive tools to enable early diagnosis and intervention. By leveraging data-driven approaches, the project aims to uncover patterns, risk factors, and potential predictive markers linked to heart diseases, facilitating preventive healthcare strategies. For this report, we will mainly focus on 3 questions. First, What are the primary risk factors associated with the presence of heart disease in the dataset? Second, Can we accurately predict the likelihood of heart disease based on demographic and clinical attributes? And last one is are there any interactions or nonlinear relationships between different risk factors and the incidence of heart disease?

II. Summary of the data.

This dataset from Kaggle labeled as Heart Disease Dataset has 13 different predict variables. Every entry includes details such as age, which is a continuous variable, sex, which is a categorical variable and basic patients' heat health information like blood pressure and so on. The variable to indicate the presence of heart disease is labeled as “target”, with 0 denoting no disease and 1 indicating having heart disease. It comprises both categorical and numerical features, making it suitable for exploratory analysis and predictive modeling. Initially, we assess the data by determining the number of entries and conducting data cleansing. Once this preliminary step is completed, we leverage the refined dataset for our analysis. Additionally, we will divide our data into two sections. The training data and the test data are the two components that aid in the model's development.

III. Exploratory Data Analysis.

i. Data analysis

There are 14 different variables including 13 predictors such as age and sex and 1 response variable which is target. There are 1025 observations in total after cleaning the data. We get the mean, range and standard deviation of each variable, these give us a foundational understanding of the dataset.

ii. Correlation

A correlation matrix shows the correlation coefficients between pairs of variables in a dataset, indicating the strength and direction of their linear relationships. Values close to +1 or -1 indicate strong positive or negative relationships, respectively, while values near 0 indicate weak or no linear relationship. It helps in understanding the relationships between variables, identifying multicollinearity, and selecting important features for model building. For instance, in this dataset, a moderate positive correlation (0.4349) between chest pain type and heart disease presence might suggest that certain chest pain types are more associated with heart disease, while a weak negative correlation (-0.2795) between sex and heart disease indicates a slight variation in heart disease likelihood based on gender. By using the correlation matrix, it helps us to select the most relevant features to do the analysis. And we can use figure 1 to see which predictor is strongest correlated to heart disease, and it is chest pain type which has the most positive correlation and oldpeak (ST depression induced by exercise relative to rest) which has the most negative correlation. So, we can mainly focus on these two variables and first we will check the distribution. We can see the result in figure 2.

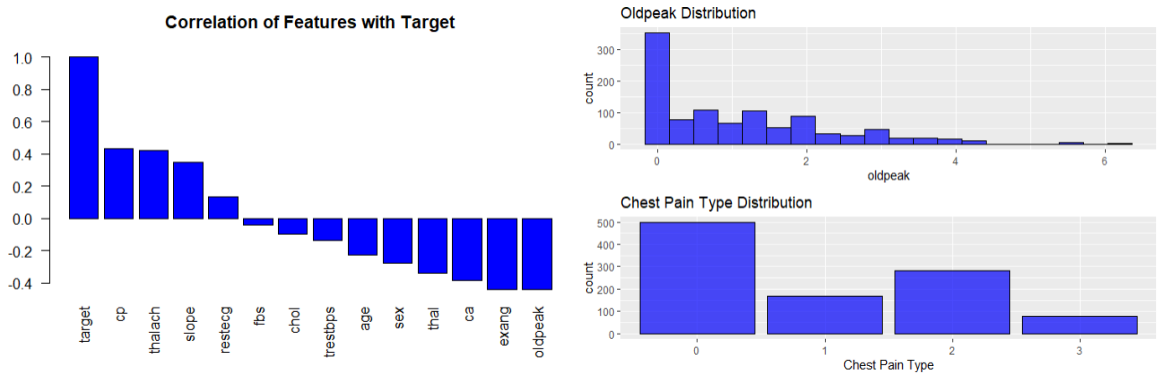


Figure 1.

iii. Visualization

a. Boxplot

From this boxplot, we can see that Cholesterol (chol) and thalach (maximum heart rate achieved) show the widest distributions and several outliers, indicating variability in these measurements among individuals. Variables like age, cp, fbs, oldpeak, ca and thal have smaller ranges and fewer outliers, suggesting more consistent values across the dataset. Overall, this boxplot provides a visual summary of the spread and central tendency of each variable in the heart disease dataset, highlighting where values are concentrated and where there are outliers.

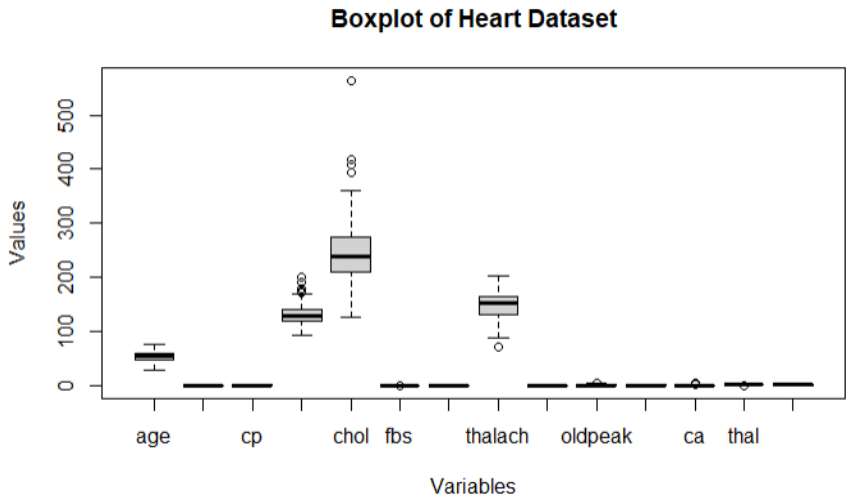


Figure 2.

b. Histogram

From the histogram, we can see that patients are predominantly middle-aged to elderly and there is a higher proportion of males. Most patients have specific types of chest pain. There are wide variations in blood pressure and cholesterol with some common ranges and for the binary variables like fbs and exang show that most patients fall into one category. These histograms provide a comprehensive overview of the distribution and frequency of various health indicators within the dataset, which can help in identifying patterns and potential risk factors related to heart disease.

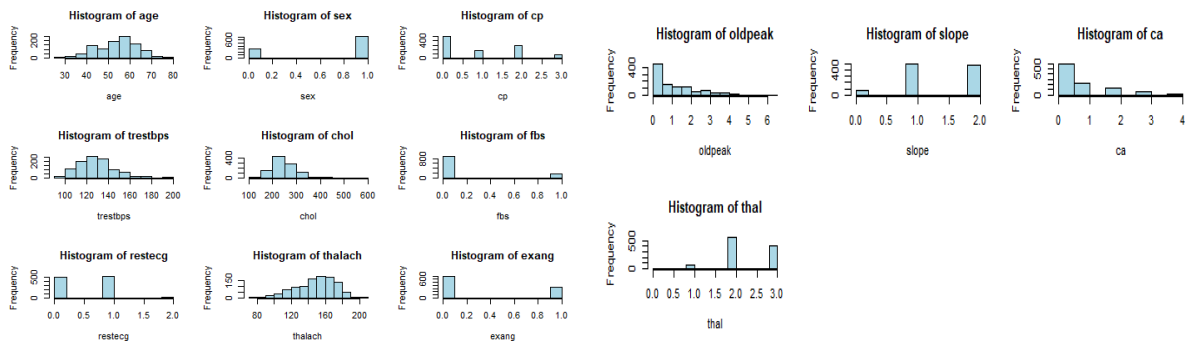


Figure 3.

c. Scatter plot

From the scatter, we can see the younger individuals tend to achieve higher maximum heart rates compared to older individuals. This trend is generally expected due to the physiological decline in maximum heart rate with age. The color differentiation in the scatter plot helps to visualize how the presence of heart disease (target variable) correlates with age and maximum heart rate. If people with heart disease cluster differently compared to those without heart disease, it could indicate a pattern such as people with heart disease might be more prevalent in older age groups. Those with heart disease might achieve lower maximum heart rates compared to those without.

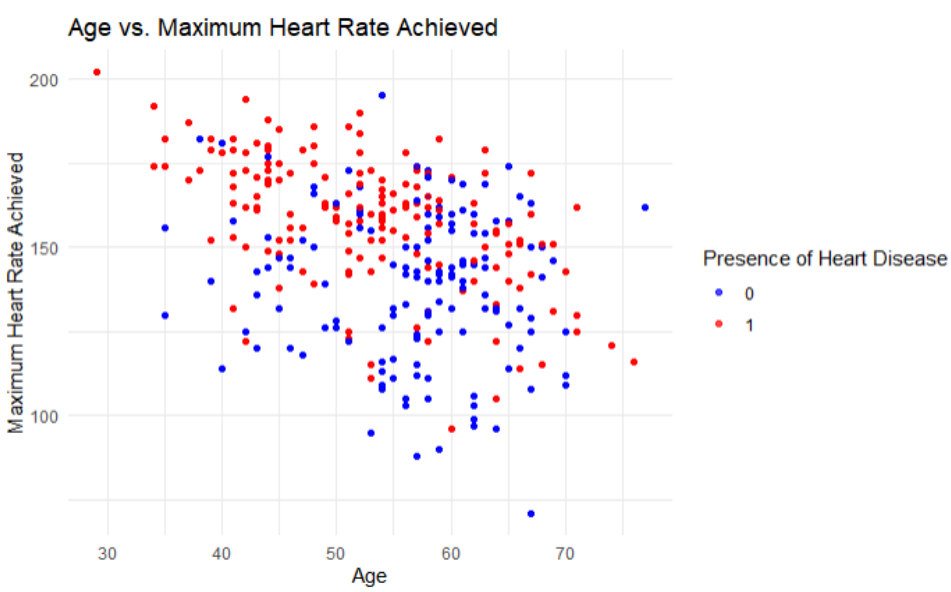


Figure 4.

IV. Model Selection.

i. Logistic regression

We will build and evaluate different machine learning models to predict heart disease such as logistic regression. And we can use the confusion matrix and ROC-AUC to evaluate the model. Logistic regression is chosen for its simplicity and effectiveness in binary classification tasks. This model estimates the probability that a given input belongs to a certain class such as the presence or absence of heart disease by fitting the input data to a logistic curve. The ROC curve is a plot of the true positive rate (sensitivity) against the false positive rate (1-specificity) for different threshold values. The area under the ROC curve (AUC) provides a single measure of overall model performance. A model with an AUC of 1.0 is perfect, whereas an AUC of 0.5 indicates a model that performs no better than random guessing. From the confusion matrix table, we can get that the accuracy for this model is 0.8235 which is not bad. For our logistic regression model, the AUC is 0.8975, indicating a high level of accuracy. This means the model is effective in distinguishing between patients with and without heart disease.

Table 1. Confusion matrix

Prediction / Ture	0	1
0	79	16
1	20	89

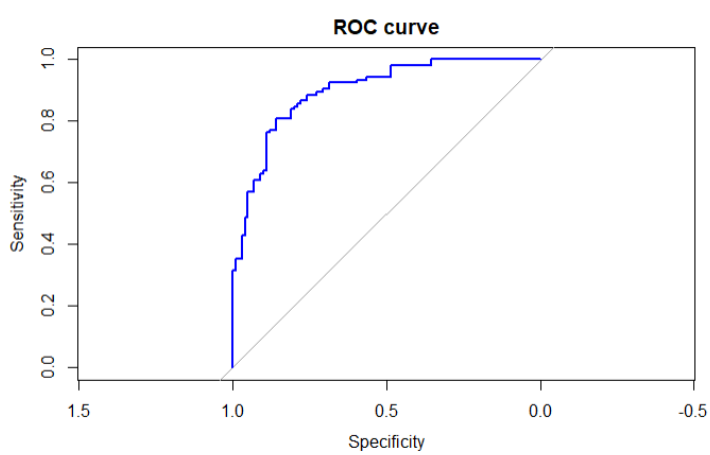


Figure 5.

ii. K_NN

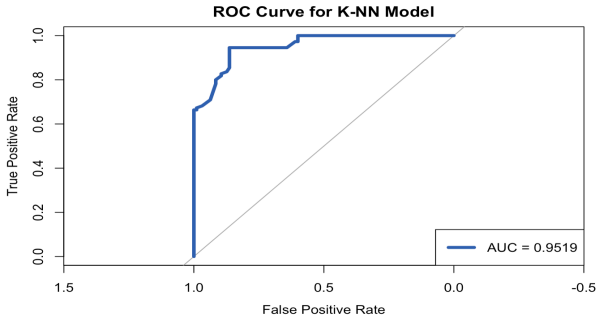


Figure 6.

Another model we decided to evaluate our data on was the K-nearest neighbor (K-NN). K-NN doesn't assume any specific form for the underlying data shape or distribution which makes it flexible for different types of data. This is great in this case the many predictor variables in our heart disease data including many complex relationships that are not easily captured by other models like logistic regression. Also, K-NN performs better on small or medium datasets which fits with the heart disease dataset having only 1025 entries and we ended up using 80% for the training data. After running K-NN, we found that The model correctly classified 85.85% of the presence of heart disease in our test data which is slightly better than the logistic regression model. Our Sensitivity and Specificity values were 0.8947 and 0.8273 respectively which meant the model correctly identified 89.47% of the actual positive cases and 82.73% of the actual negative cases. Another important value to look at is AUC which is the area under the roc curve which is about 0.952 which is very high, close to 1. This indicates that the K-NN model is excellent at distinguishing between the positive and negative classes and better overall model performance in classification.

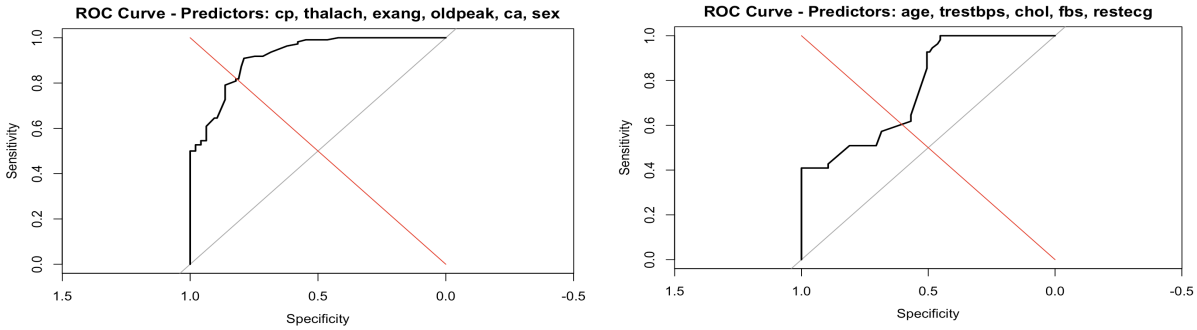
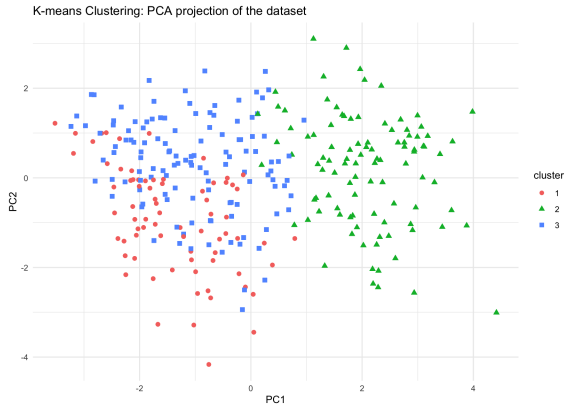
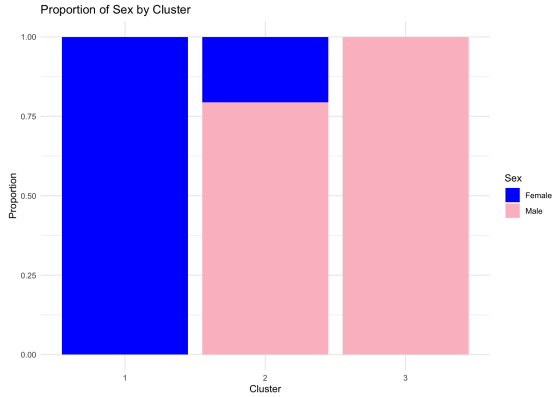


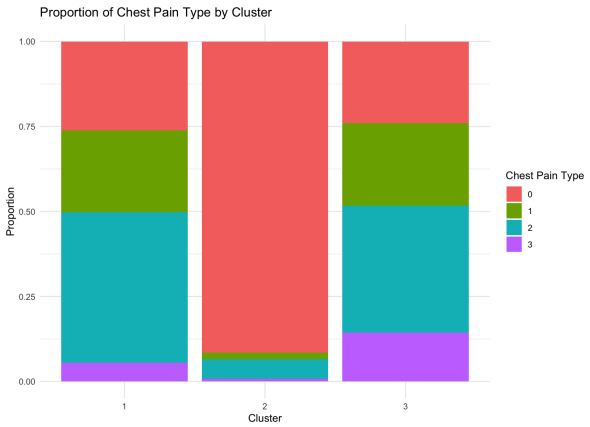
Figure 7.

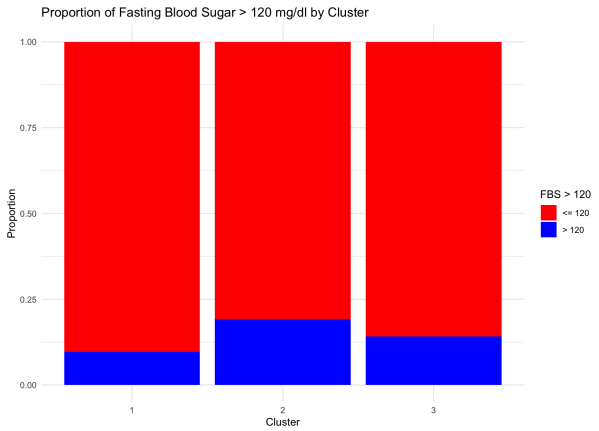
To help us identify and prove which predictor variables are the best. I decided to run 2 different K-NN models where one only consider the “better” predictors which I took from our correlation plot that included the variables chest pain (cp), maximum heart rate achieved (thalach), exercise induced angina (exang), ST depression from exercise (oldpeak), number of major vessels (ca) and sex. The model using the best predictors has an accuracy of about 84.88% which is slightly lower than using all predictor variables but this goes to show that only using about half of the total predictors we have is almost enough to diagnose heart disease. The other K-NN model considered the remaining predictors that were found not to be as significant against the target variable and had an accuracy of 61.46%. This is little over 20% worse than the other two models showing the significant drop of performance in these variables.

K-means Clustering









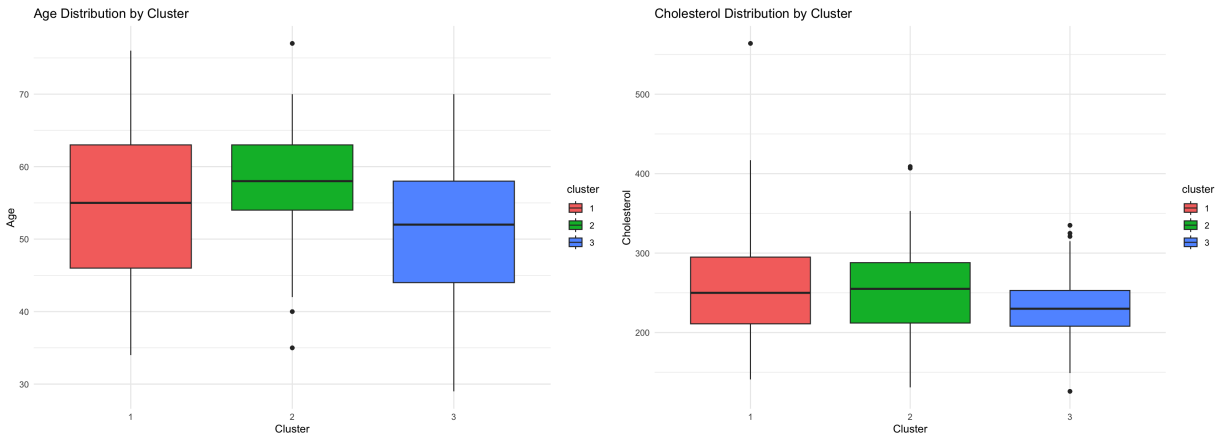


Figure 8.

We applied K-means clustering to the dataset of heart disease patients to identify distinct groups based on various health indicators. The clustering process resulted in three distinct clusters, each with unique characteristics that provide valuable insights into the different patient profiles.

Cluster 1 consists of 237 patients, all of whom are female. This cluster can be characterized by a moderate risk profile. The average age of patients in this cluster is 54.8 years. These patients generally experience typical or atypical angina, as suggested by an average chest pain type (cp) value of 1.29. They have an average resting blood pressure (trestbps) of 128.99 mm Hg and a relatively high average serum cholesterol level (chol) of 257.72 mg/dl. Only 9.7% of these patients have fasting blood sugar (fbs) levels above 120 mg/dl. The average maximum heart rate achieved (thalach) is 154.53 beats per minute, and 10.1% experience exercise-induced angina (exang). The average ST depression induced by exercise relative to rest (oldpeak) is 0.53 mm, and the average slope of the peak exercise ST segment (slope) is 1.59. On average, these patients have 0.33 major vessels (ca) colored by fluoroscopy, indicating fewer blockages. The average thalassemia (thal) value is 2.00, suggesting a mix of normal and fixed defects. A significant finding is that 95.4% of the patients in this cluster are diagnosed with heart disease, indicating a high prevalence within this group which we can see through the variables as well.

Cluster 2 includes 364 patients and represents a mixed-sex group, with 75 females and 289 males. This cluster is characterized by a low risk profile. The patients in this cluster have an average age of 57.8 years. They are more likely to be asymptomatic or have typical angina, as indicated by an average cp value of 0.16. These patients have a

higher average resting blood pressure of 135.85 mm Hg and an average serum cholesterol level of 254.00 mg/dl. In this cluster, 19.2% of patients have fasting blood sugar levels exceeding 120 mg/dl. The average maximum heart rate achieved is lower at 131.05 beats per minute, and a significant proportion, 76.6%, experience exercise-induced angina. The average ST depression induced by exercise relative to rest is 1.90 mm, and the average slope of the peak exercise ST segment is 1.01. These patients have an average of 1.25 major vessels colored by fluoroscopy, indicating more extensive arterial blockages. The average thalassemia value is 2.55, reflecting a higher prevalence of fixed and reversible defects. Notably, only 4.1% of patients in this cluster are diagnosed with heart disease, suggesting a lower prevalence compared to the other clusters.

Cluster 3 is composed of 424 patients, all of whom are male. This cluster is characterized by a high risk profile. The average age of patients in this cluster is 51.3 years, making them the youngest among the three clusters. These patients often experience typical angina or non-anginal pain, as suggested by an average cp value of 1.42. They have an average resting blood pressure of 129.43 mm Hg and a lower average serum cholesterol level of 232.58 mg/dl. In this cluster, 14.1% of patients have fasting blood sugar levels above 120 mg/dl. The average maximum heart rate achieved is higher at 161.59 beats per minute, and only 9.9% experience exercise-induced angina. The average ST depression induced by exercise relative to rest is 0.66 mm, and the average slope of the peak exercise ST segment is 1.60. On average, these patients have 0.56 major vessels colored by fluoroscopy, indicating fewer blockages. The average thalassemia value is 2.31, reflecting a mix of normal, fixed, and reversible defects. A high proportion, 67.2%, of patients in this cluster are diagnosed with heart disease, indicating a significant risk within this group, although it has a significantly lower proportion than cluster 1.

The clustering results reveal significant differences based on sex, age, and various health indicators. While it is acceptable for clusters to be separated by sex if it reflects the underlying structure of the data, it is essential to consider other variables to ensure comprehensive and meaningful clustering. If sex strongly influences the clusters, additional analyses or subgroup clustering may be necessary to capture more nuanced patterns and provide a more detailed understanding of the patient groups. Overall, these findings highlight the importance of personalized medical approaches based on patient profiles to improve heart disease management and treatment outcomes.

V. Conclusion.

Our exploration of the Heart Disease Dataset from Kaggle aimed to identify primary risk factors, predict heart disease likelihood, and explore interactions among risk factors. Key findings highlighted that chest pain type, maximum heart rate achieved (thalach), and oldpeak (ST depression) are significantly correlated with heart disease presence, with chest pain type being the strongest predictor. Logistic regression and KNN models were tested, achieving AUCs of 0.8975 and 0.952, respectively, with KNN showing superior discriminatory ability. These insights affirm the complexity of heart disease and support effective prediction based on demographic and clinical attributes. This report helps us to answer the questions in the beginning as well.

VI. Appendix.

```
heart = read.csv("E:/University/sta141a/final proj/heart.csv")
library(tidyverse)
library(caret)
library(randomForest)
library(e1071)
library(nnet)
library(pROC)
library(ggplot2)
library(gridExtra)

str(heart)
summary(heart)
colSums(is.na(heart))
heart = na.omit(heart)
cor_matrix = cor(heart)
print(cor_matrix)
oldpeak = ggplot(heart, aes(x = oldpeak)) +
  geom_histogram(bins = 20, fill = 'blue', color = 'black', alpha = 0.7) +
  ggtitle('Oldpeak Distribution')

cp = ggplot(heart, aes(x = factor(cp))) +
  geom_bar(fill = 'blue', color = 'black', alpha = 0.7) +
  ggtitle('Chest Pain Type Distribution') +
  xlab('Chest Pain Type')

cor_target = cor_matrix[,'target']
cor_target = sort(cor_target, decreasing = TRUE)
barplot(cor_target, main = "Correlation of Features with Target", col = 'blue', las = 2)

grid.arrange(oldpeak, cp, nrow = 2)
set.seed(1)
heart$target = as.factor(heart$target)
```

```
index = createDataPartition(heart$target, p = 0.8, list = FALSE)
train_heart = heart[index, ]
test_heart = heart[-index, ]

preProc = preProcess(train_heart[, -ncol(train_heart)], method = c("center", "scale"))
train_heart_scaled = predict(preProc, train_heart[, -ncol(train_heart)])
test_heart_scaled = predict(preProc, test_heart[, -ncol(test_heart)])

train_heart = cbind(train_heart_scaled, target = train_heart$target)
test_heart = cbind(test_heart_scaled, target = test_heart$target)

log_reg = train(target ~ ., data = train_heart, method = 'glm', family = 'binomial')
log_reg
summary(log_reg)

log_reg_preds = predict(log_reg, test_heart)
log_reg_prob = predict(log_reg, test_heart, type = "prob")[,2]
conf_matrix = confusionMatrix(log_reg_preds, test_heart$target)
conf_matrix
roc_curve = roc(test_heart$target, log_reg_prob)
auc = auc(roc_curve)

plot(roc_curve, col = "blue", main = "ROC curve")
auc

boxplot(heart, main="Boxplot of Heart Dataset", xlab="Variables", ylab="Values")
ggplot(heart, aes(x = age, y = thalach, color = factor(target))) +
  geom_point(alpha = 0.7) +
  scale_color_manual(values = c("blue", "red"), name = "Presence of Heart Disease") +
  labs(title = "Age vs. Maximum Heart Rate Achieved",
       x = "Age",
       y = "Maximum Heart Rate Achieved") +
  theme_minimal()
```

```r
# Splitting the data into train and test sets
set.seed(123)
train_indices <- sample(1:nrow(heart), 0.8 * nrow(heart))
train_data <- heart[train_indices, ]
test_data <- heart[-train_indices, ]

# Preprocess the data: center and scale
preProc <- preProcess(train_data[, -ncol(train_data)], method = c("center", "scale"))
train_data_scaled <- predict(preProc, train_data[, -ncol(train_data)])
test_data_scaled <- predict(preProc, test_data[, -ncol(test_data)])

# Combine scaled features with target variable
train_data <- cbind(train_data_scaled, target = train_data$target)
test_data <- cbind(test_data_scaled, target = test_data$target)

# Train K-NN model
set.seed(123)
knn_model <- train(target ~ ., data = train_data, method = "knn", trControl = trainControl(method = "cv", number =
10), tuneLength = 10)

# Print the best k
print(knn_model)

# Predicting on the test set
knn_predictions <- predict(knn_model, newdata = test_data)

# Confusion matrix
confusion_matrix <- confusionMatrix(knn_predictions, test_data$target)
print(confusion_matrix)

# Accuracy
accuracy <- mean(knn_predictions == test_data$target)
print(paste("Accuracy:", accuracy))
```

```r
# ROC Curve and AUC
knn_probs <- predict(knn_model, newdata = test_data, type = "prob")[,2]
knn_roc <- roc(test_data$target, knn_probs)

# Plot ROC Curve with labels
plot(knn_roc, main = "ROC Curve for K-NN Model", col = "blue", lwd = 4,
    xlab = "False Positive Rate", ylab = "True Positive Rate")
legend("bottomright", legend = paste("AUC =", round(auc(knn_roc), 4)), col = "blue", lwd = 4)

# Print AUC
print(paste("AUC:", auc(knn_roc)))

perform_knn <- function(predictors, data) {
  # Subset the dataset to include only the selected predictors and the target variable
  predictors_data <- data[, c(predictors, "target")]

  # Splitting the data into train and test sets
  set.seed(123)
  train_indices <- sample(1:nrow(predictors_data), 0.8 * nrow(predictors_data))  # 80% for training
  train_data <- predictors_data[train_indices, ]
  test_data <- predictors_data[-train_indices, ]

  # Train the K-NN model using the selected predictors
  set.seed(123)
  knn_model <- train(target ~ ., data = train_data, method = "knn")

  # Predicting on the test set
  predictions <- predict(knn_model, newdata = test_data)

  # Confusion matrix
  cm <- confusionMatrix(predictions, test_data$target)
  print(cm)
```

```r
  # Accuracy
  accuracy <- mean(predictions == test_data$target)
  print(paste("Accuracy:", accuracy))

  # AUC
  probs <- predict(knn_model, newdata = test_data, type = "prob")[,2]
  roc <- roc(test_data$target, probs)
  print(paste("AUC:", auc(roc)))

  # Plot ROC curve
  plot(roc, main = paste("ROC Curve - Predictors:", paste(predictors, collapse = ", ")))
  lines(x = c(0, 1), y = c(0, 1), col = "red")  # Add diagonal reference line
}

# Call the function for the best predictors
perform_knn(c("cp", "thalach", "exang", "oldpeak", "ca", "sex"), heart)

# Call the function for the best predictors
perform_knn(c("age", "trestbps", "chol", "fbs", "restecg"), heart)

# Ensure only numeric columns are being used for clustering
numeric_cols <- sapply(heart, is.numeric)
numeric_data <- heart[, numeric_cols]

# Standardize the numeric data
numeric_data_scaled <- as.data.frame(scale(numeric_data))

# Determine the optimal number of clusters using the Elbow method
wss <- (nrow(numeric_data_scaled) - 1) * sum(apply(numeric_data_scaled, 2, var))
for (i in 2:15) wss[i] <- sum(kmeans(numeric_data_scaled, centers = i)$withinss)
plot(1:15, wss, type = "b", xlab = "Number of Clusters", ylab = "Within groups sum of squares")

# Perform K-means clustering with the chosen number of clusters
```

```r
set.seed(42)
kmeans_result <- kmeans(numeric_data_scaled, centers = 3, nstart = 25)

# Add cluster assignments to the original dataset
heart$cluster <- as.factor(kmeans_result$cluster)

# Visualize the clusters
pca <- prcomp(numeric_data_scaled, center = TRUE, scale. = TRUE)
pca_data <- as.data.frame(pca$x[, 1:2])
pca_data$cluster <- heart$cluster

ggplot(pca_data, aes(x = PC1, y = PC2, color = cluster, shape = cluster)) +
  geom_point(size = 2) +
  labs(title = "K-means Clustering: PCA projection of the dataset",
       x = "PC1",
       y = "PC2") +
  theme_minimal()

# Summarize the clusters
summary(heart$cluster)
aggregate(numeric_data, by = list(heart$cluster), FUN = mean)

# Proportion of males and females by cluster
ggplot(heart, aes(x = cluster, fill = as.factor(sex))) +
  geom_bar(position = "fill") +
  labs(title = "Proportion of Sex by Cluster", x = "Cluster", y = "Proportion", fill = "Sex") +
  scale_fill_manual(values = c("blue", "pink"), labels = c("Female", "Male")) +
  theme_minimal()

# Proportion of chest pain types by cluster
ggplot(heart, aes(x = cluster, fill = as.factor(cp))) +
  geom_bar(position = "fill") +
  labs(title = "Proportion of Chest Pain Type by Cluster", x = "Cluster", y = "Proportion", fill = "Chest Pain Type") +
```

```
  theme_minimal()

# Proportion of fasting blood sugar > 120 mg/dl by cluster
ggplot(heart, aes(x = cluster, fill = as.factor(fbs))) +
  geom_bar(position = "fill") +
  labs(title = "Proportion of Fasting Blood Sugar > 120 mg/dl by Cluster", x = "Cluster", y = "Proportion", fill =
"FBS > 120") +
  scale_fill_manual(values = c("red", "blue"), labels = c("<= 120", "> 120")) +
  theme_minimal()

# Boxplot of age by cluster
ggplot(heart, aes(x = cluster, y = age, fill = cluster)) +
  geom_boxplot() +
  labs(title = "Age Distribution by Cluster", x = "Cluster", y = "Age") +
  theme_minimal()

# Boxplot of cholesterol by cluster
ggplot(heart, aes(x = cluster, y = chol, fill = cluster)) +
  geom_boxplot() +
  labs(title = "Cholesterol Distribution by Cluster", x = "Cluster", y = "Cholesterol") +
  theme_minimal()

# Summary of the sex variable within each cluster
table(heart$cluster, heart$sex)
```